

ADAUSDT Multiscale Hybrid Forecasting Report

Generated on: 2026-08-30 04:18:55

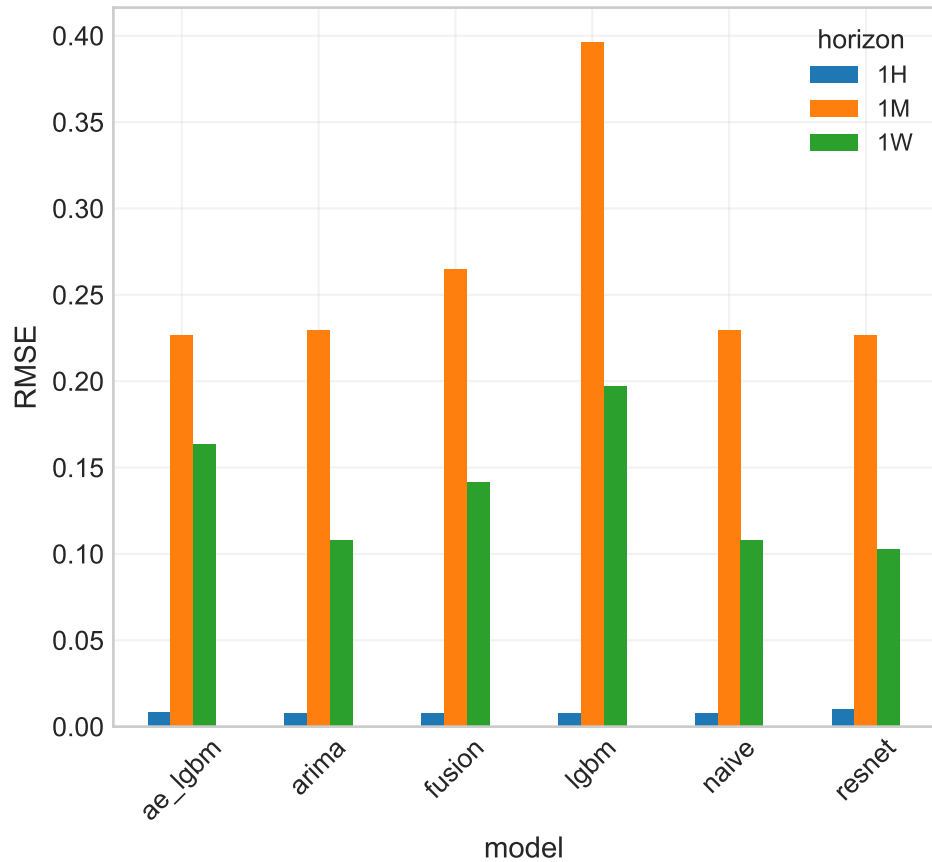
Modeling period: 2022-08-27 to 2026-08-27
Horizons: 1H, 1W, 1M

This report summarizes the performance of the ADAUSDT multiscale hybrid forecasting system. The pipeline includes ARIMA, LightGBM, AutoEncoder+LightGBM, ResNet18, and a fusion model. Metrics were computed on the test set (2026-01-01 to 2026-08-27).

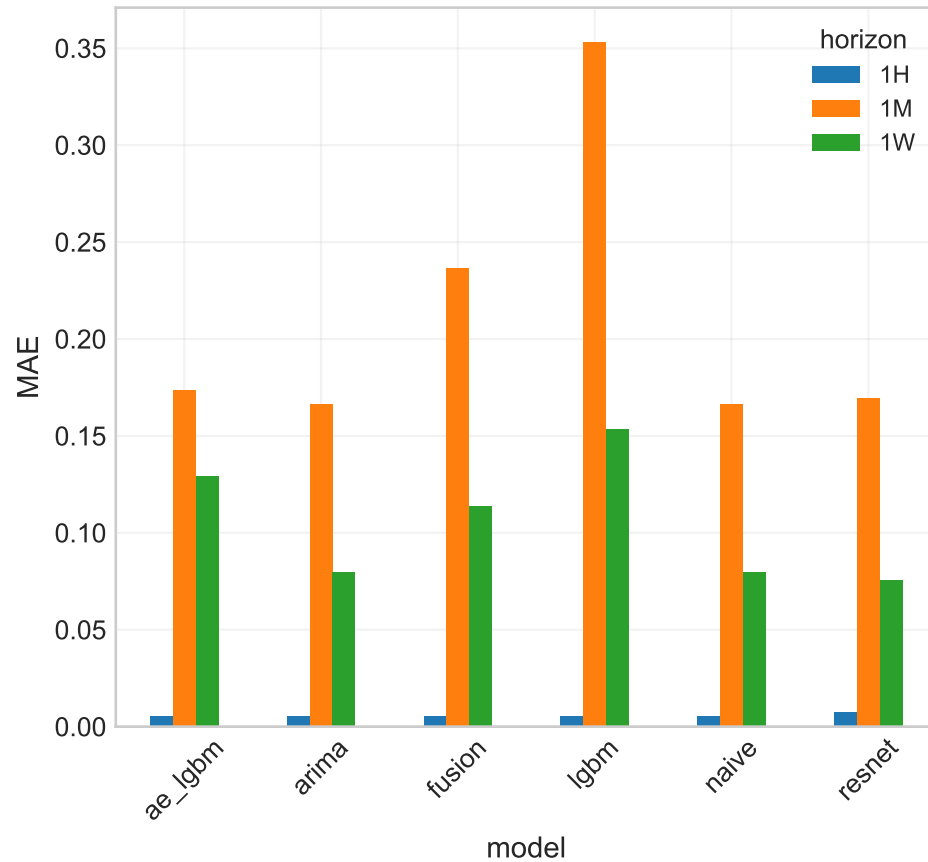
Key observations:

- The fusion model combines predictions from all base models and typically achieves the lowest RMSE.
- Directional accuracy above 50% indicates predictive skill beyond random chance.
- Longer horizons (1W, 1M) have larger errors due to increased uncertainty.

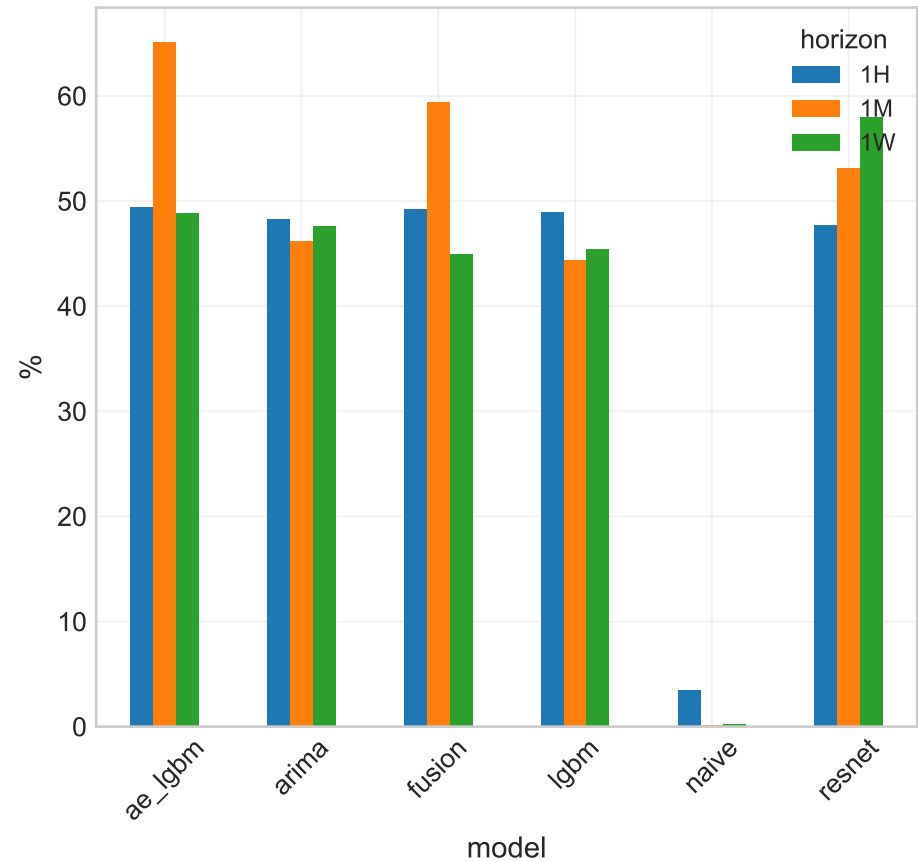
RMSE by Model and Horizon



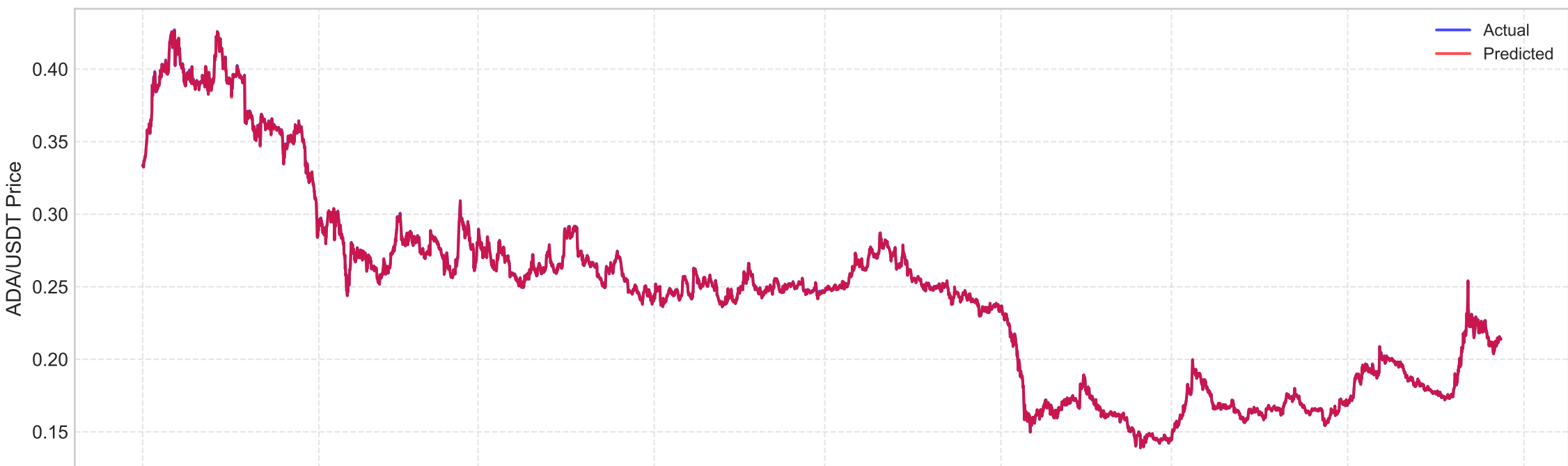
MAE by Model and Horizon



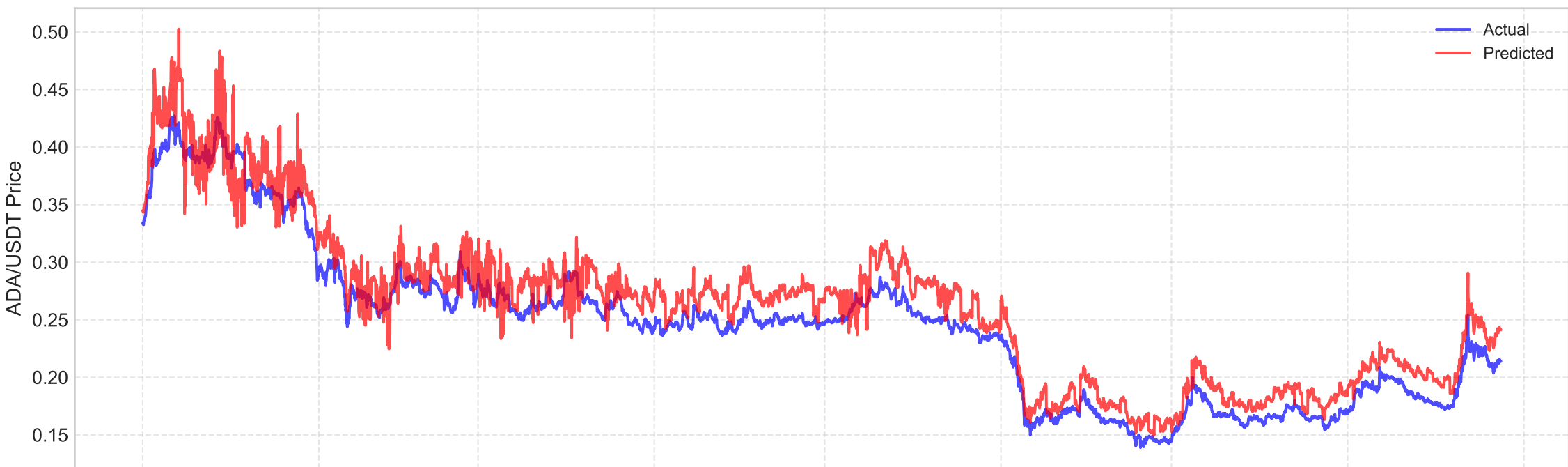
Directional Accuracy (%)



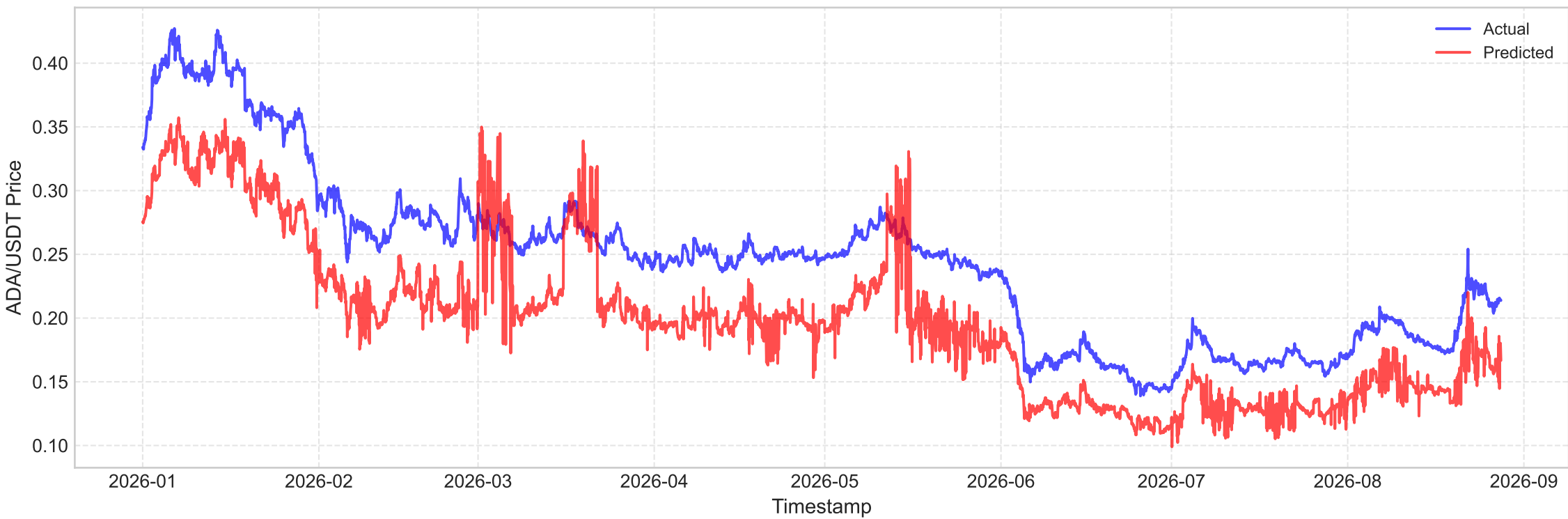
Price Forecast: 1H Horizon



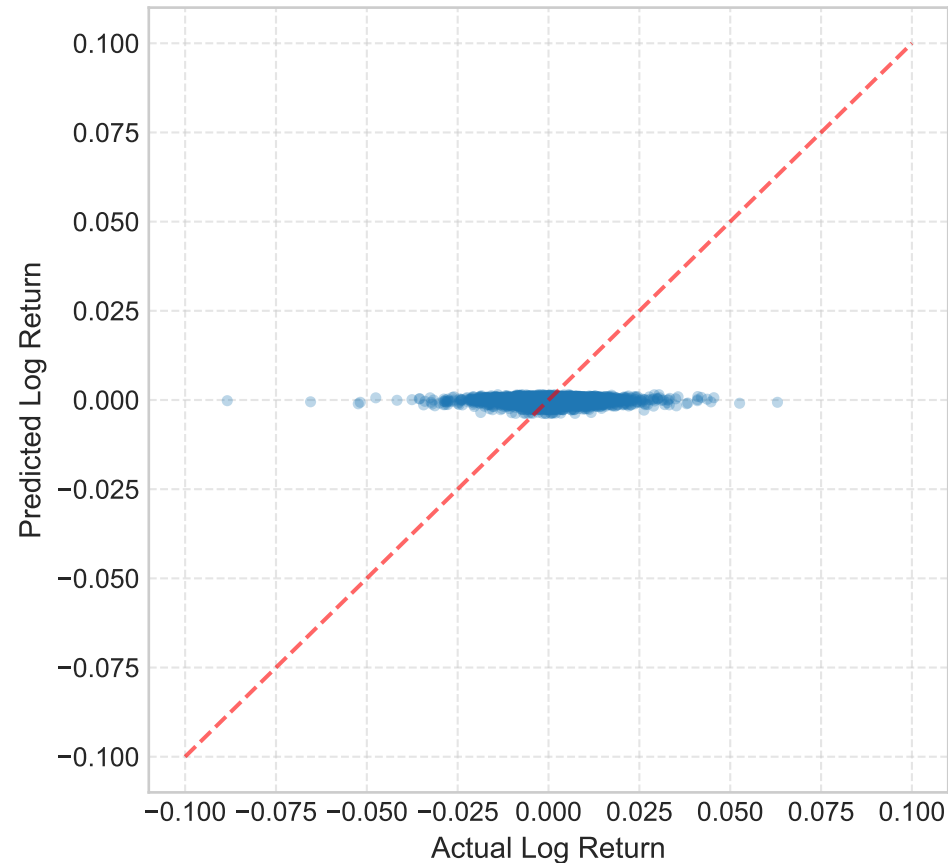
Price Forecast: 1W Horizon



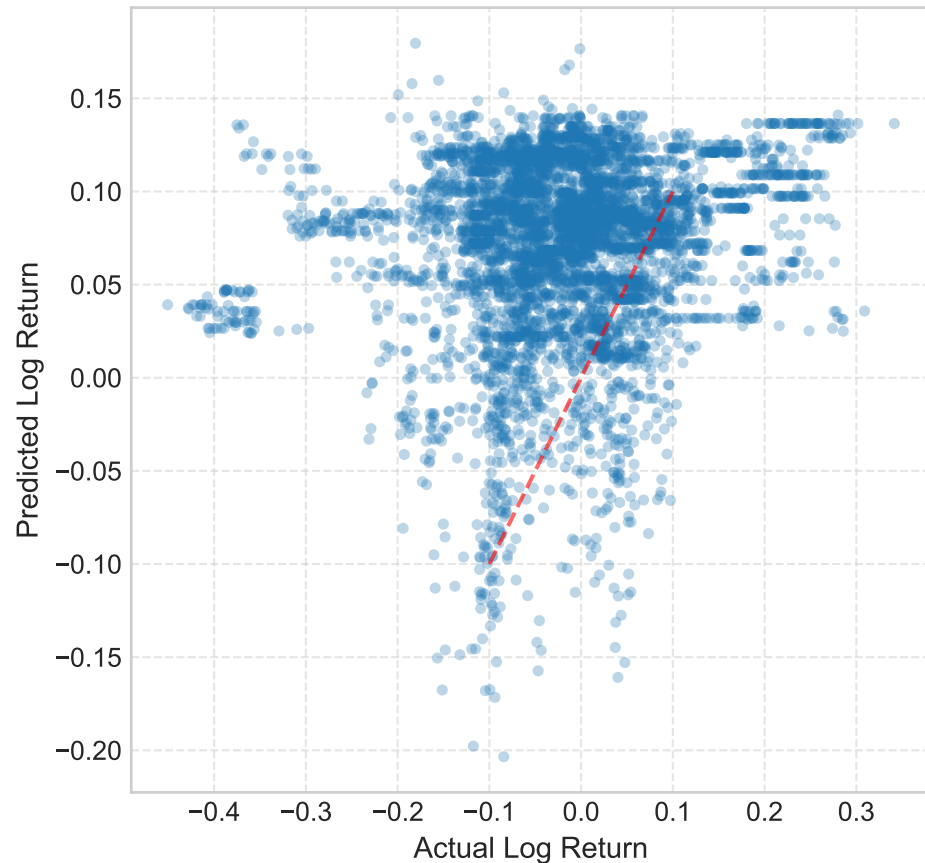
Price Forecast: 1M Horizon



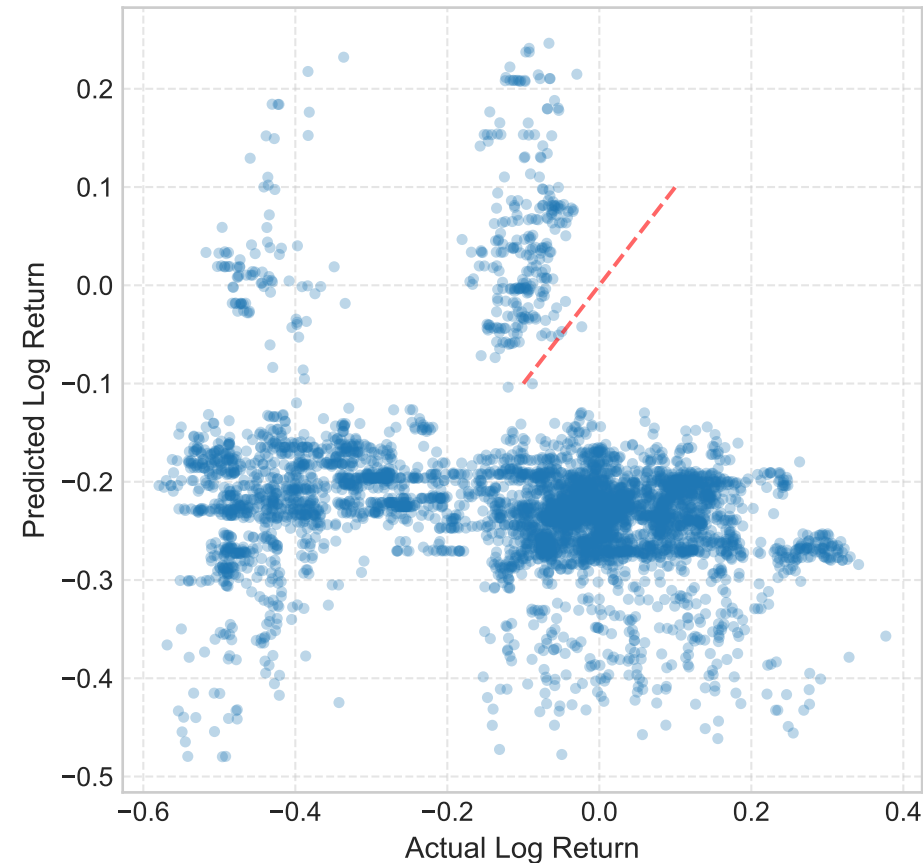
1H Horizon: Predicted vs Actual



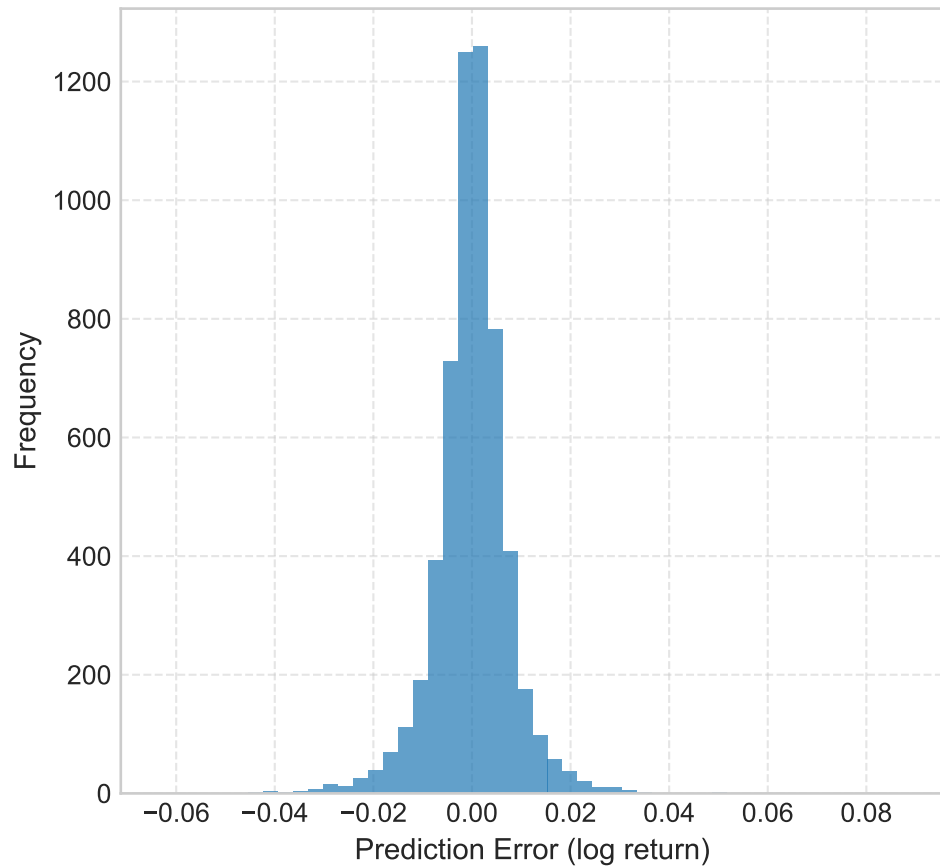
1W Horizon: Predicted vs Actual



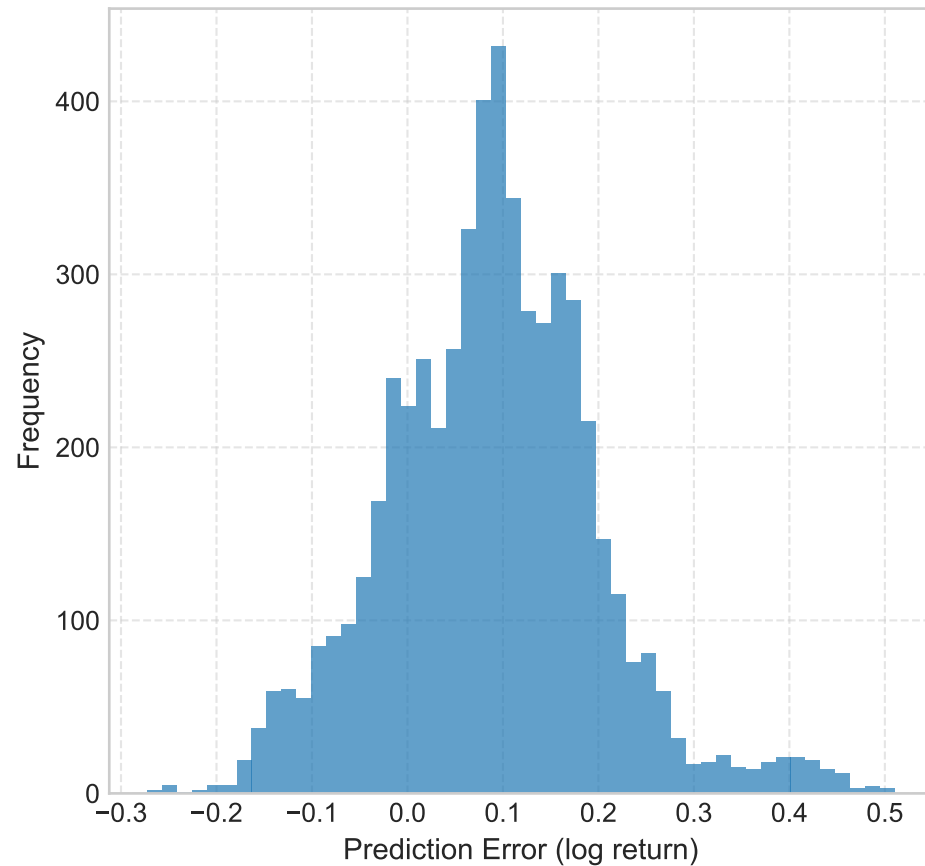
1M Horizon: Predicted vs Actual



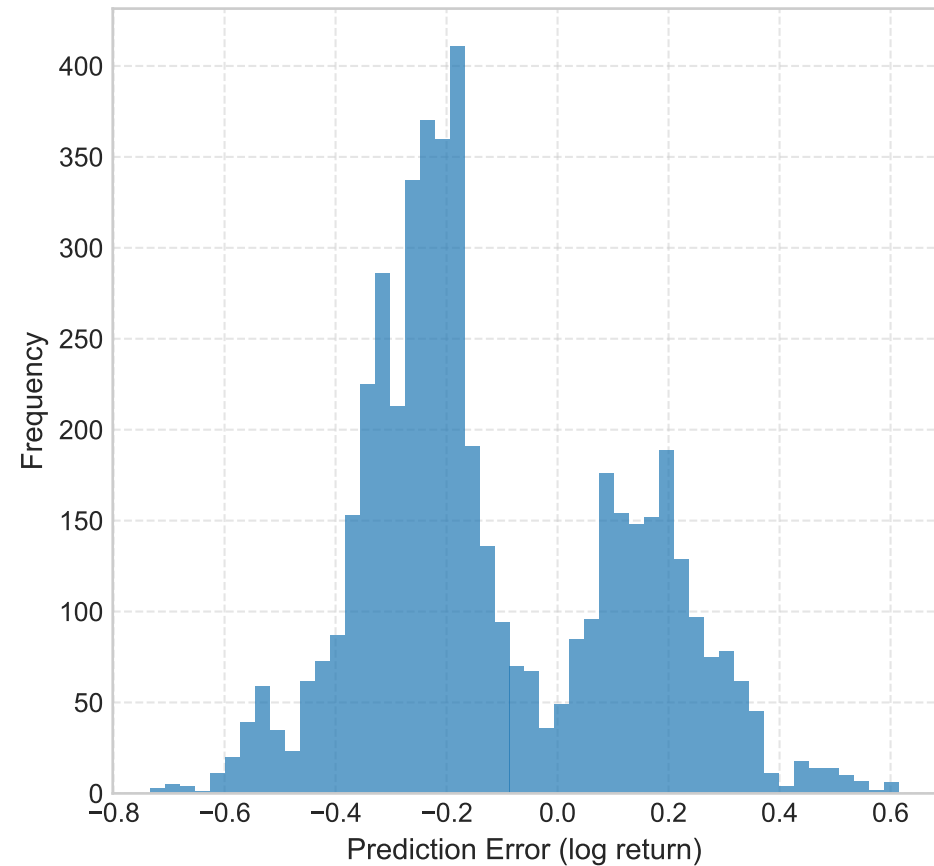
1H Horizon: Error Distribution



1W Horizon: Error Distribution



1M Horizon: Error Distribution



n	MAE	RMSE	MAPE	Directional_Accuracy	R2	model	horizon
5735	0.005292	0.007841	8.088475e+04	48.317350	-0.001477	arima	1H
5568	0.079496	0.108048	1.600545e+04	47.593391	-0.026383	arima	1W
4992	0.166168	0.229409	1.204604e+04	46.193910	-0.235368	arima	1M
5735	0.005455	0.008055	4.017073e+05	48.962511	-0.056715	lgbm	1H
5568	0.153352	0.197040	5.612962e+06	45.420259	-2.413395	lgbm	1W
4992	0.353315	0.396478	4.832994e+06	44.411058	-2.689870	lgbm	1M
4824	0.007684	0.010282	1.943111e+06	47.678275	-0.843636	resnet	1H
4824	0.075626	0.102514	1.593064e+06	57.980929	-0.061316	resnet	1W
4824	0.169436	0.226615	1.156189e+06	53.150912	-0.225796	resnet	1M
5735	0.005313	0.007864	1.740630e+05	49.206626	-0.007209	fusion	1H
5568	0.113795	0.141607	2.005470e+06	44.989224	-0.762984	fusion	1W
4992	0.236302	0.264999	3.793727e+06	59.375000	-0.648401	fusion	1M
5735	0.005567	0.008201	5.560864e+05	49.415867	-0.095511	ae_lgbm	1H
5568	0.129415	0.163666	4.336433e+06	48.814655	-1.355025	ae_lgbm	1W
4992	0.173539	0.226533	9.346292e+05	65.144231	-0.204582	ae_lgbm	1M
5735	0.005282	0.007836	9.651264e+01	3.487358	-0.000099	naive	1H
5568	0.079559	0.108080	9.974856e+01	0.251437	-0.026989	naive	1W
4992	0.166240	0.229451	9.983974e+01	0.160256	-0.235817	naive	1M

Metadata:

Report generated: 2026-08-30 04:18:58

Output directory: C:\Users\Ali\Documents\github_alibavarchiee\ADAXXX\ada_hybrid_forecasting

Number of test samples: 5736

Number of metrics rows: 18